PROJECT PLANNING PHASE

Project Title: ComicCraft – AI Comic Story Creator using Gemini Models

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1. Introduction

ComicCraft is an AI-powered comic story creation project that helps users turn a simple idea or prompt into a short comic. Using Google Gemini models, the system can help develop a story, organize it into scenes or panels, create character and dialogue descriptions, and prepare prompts for comic illustrations. The goal is to make comic creation easier, faster, and more accessible for students and beginners.

2. Problem Statement

Creating a comic traditionally requires story-writing skills, scene planning, dialogue writing, and illustration work. Beginners may find it difficult to connect these tasks and maintain consistency between scenes. ComicCraft aims to simplify this process by providing AI-assisted story development and a structured panel-by-panel output.

3. Objectives

- Generate a short comic story from a user-provided topic or prompt.
- Divide the story into a sequence of scenes or comic panels.
- Create character descriptions, narration, and dialogue for each panel.
- Produce clear image prompts that can be used to create comic illustrations.
- Allow users to review, edit, and regenerate story content.

- Provide a simple and beginner-friendly user interface.

4. Proposed System

The proposed system accepts a topic, genre, target audience, and preferred number of scenes. The Gemini model processes the prompt and returns a structured story. The application displays the story as panels, including scene descriptions, dialogue, and illustration prompts. Users can revise the prompt or edit the generated content before saving it.

5. Scope of the Project

- Create short comics for educational, entertaining, or storytelling purposes.
- Support a range of genres such as adventure, fantasy, mystery, and school life.
- Generate a configurable number of scenes (for example, 3–6 panels).
- Provide text-based story and image-prompt output; image generation may be added if supported by the selected tools.
- Keep the user in control of final edits and the finished comic.

6. Main Modules

Module	Description
User Input Module	Collects the topic, genre, audience, scene count, and optional character details.
Prompt Builder	Converts user choices into clear instructions for the Gemini model.
Story Generation Module	Generates the story outline, scene sequence, and narrative flow.
Panel and Dialogue Module	Creates panel descriptions, character actions, captions, and dialogue.
Illustration Prompt Module	Creates visual prompts describing characters, setting, mood, and composition.
Preview and Editing Module	Displays the generated comic plan and allows users to revise content.
Export Module	Provides a way to copy or save the comic plan for later use.

7. Tools and Technologies

Technology	Planned Use
AI Model	Google Gemini models via an available Gemini API or supported interface
Frontend	HTML, CSS, and JavaScript, or a suitable app-building framework
Backend (if required)	Python / Node.js to manage requests and application logic
Output	Comic script, panel descriptions, dialogue, and illustration prompts
Development Tools	Code editor, browser, and version-control tool (optional)

Note: The exact Gemini model and API access will be selected during implementation. API keys must be kept private and should not be placed in public client-side code.

8. Basic Workflow

1. The user enters a topic and selects the genre and number of panels.
2. The system builds a structured prompt from the selected options.
3. Gemini generates a story outline and panel-by-panel content.
4. The application formats the response into a comic storyboard.
5. The user reviews and edits the content.
6. The final comic plan is copied or exported.

9. Functional Requirements

- The system shall accept a text prompt from the user.
- The system shall allow selection of genre and panel count.
- The system shall generate a coherent story with a beginning, middle, and ending.
- Each panel shall include a scene description and relevant dialogue or caption.
- The user shall be able to edit or regenerate generated content.
- The system shall show an understandable error message if generation fails.

10. Non-Functional Requirements

- Usability: The interface should be simple and easy to understand.

- Performance: Story generation should complete within a reasonable time, depending on model response.
- Reliability: The application should handle empty prompts and service errors gracefully.
- Privacy: Avoid collecting unnecessary personal information.
- Security: Protect API credentials and validate user input.
- Quality: Generated stories should be readable, age-appropriate, and logically connected.

11. Feasibility Study

Technical Feasibility: The project can be developed using common web technologies and an available Gemini model/API.

Economic Feasibility: A basic prototype can be built with low-cost or free development tools. API usage may incur charges depending on the service plan.

Operational Feasibility: The system is designed for students, teachers, and beginner storytellers, with a simple prompt-and-preview workflow.

12. Tentative Project Schedule

Timeline	Task
Week 1	Requirement collection and project planning
Week 2	UI design and prompt structure
Week 3	Gemini integration and story generation
Week 4	Panel formatting, editing, and export
Week 5	Testing, corrections, and documentation

13. Expected Outcome

The expected outcome is a working prototype that converts a user's idea into a structured comic storyboard. It should provide a title, characters, a sequence of panels, dialogue or captions, and illustration prompts. The result will help users plan and create comics with less manual effort.

14. Risks and Mitigation

Risk	Mitigation
Inconsistent characters or story details	Include character descriptions in the prompt and review each panel.
Incorrect or unsuitable generated content	Use clear instructions and provide user review before finalizing.
API limits or connectivity issues	Show an error message and allow the user to retry.
Unexpected API costs	Set usage limits and monitor API consumption.
Privacy or key exposure	Do not include secrets in public code; minimize stored user data.

15. Conclusion

ComicCraft proposes a simple way to combine AI-assisted writing with comic storyboard planning. The planning phase defines the project objectives, scope, modules, technologies, workflow, requirements, and schedule. These plans will guide the design, development, testing, and demonstration of the ComicCraft prototype.